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BIOMEDICAL IMAGE PROCESSING FOR ENHANCED CLASSIFICATION OF MEDICINAL PLANTS USING VOTING ENSEMBLE

Nova Agustina[1,*], Candra Nur Ihsan[2], Kelik Sussolaikah[3]

[1]Department of Informatics, Universitas Teknologi Bandung, West Java, Indonesia
[2]Badan Riset dan Inovasi Nasional, Jakarta, Indonesia
[3]Department of Informatics, Universitas PGRI Madiun, East Java, Indonesia

Abstract

Indonesia harbors extensive biodiversity, including many medicinal plants used in health. Accurate and scalable identification remains challenging because of high intra class variation, strong visual similarity among species, and variable capture conditions in the field. This study evaluates three convolutional neural networks for Indonesian medicinal plant classification: ResNet18, VGG16, and EfficientNet_B0, and an average voting ensemble that aggregates their class probabilities. Using a held-out test set, ResNet18 achieves 87.00% accuracy, 91.16% precision, 87.00% recall, and 87.35% F1 score; VGG16 records 63.50% accuracy, 63.89% precision, 63.50% recall, and 61.10% F1 score; EfficientNet_B0 attains 94.00% accuracy, 94.49% precision, 94.00% recall, and 94.01% F1 score; the ensemble reaches 93.00% accuracy, 94.99% precision, 93.00% recall, and 93.22% F1 score. Confusion matrix analysis indicates that remaining errors concentrate on visually similar pairs, notably Aloe Vera versus Green Tea and Basil Leaf versus Green Tea. The findings show that EfficientNet_B0 is the strongest single backbone, while the ensemble improves precision and stability with accuracy close to the best individual model. These results demonstrate a practical pathway for reliable, field deployable recognition of medicinal plants in Indonesia. Future work will expand the dataset across seasons and regions, increase class balance, and explore stronger augmentation, focal or class weighted losses, and probability calibration. We also plan segmentation assisted pipelines, staged fine tuning at higher resolution, and lightweight deployment through pruning and quantization on mobile or embedded devices. To support reproducibility and adoption, the training pipeline and plotting scripts are made available in an open Kaggle notebook.

Keywords: Classification, Ensemble, Medicinal-plants, Voting

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I. INTRODUCTION

Indonesia is one of the most diverse countries in the world, especially in terms of medicinal plants. Traditional medicinal plants have long been used by Indonesian people for various medicinal and health purposes [1]. In fact, Indonesian medicinal plants are also exported abroad [2], which has an impact on increasing the country's foreign exchange. Medicinal plants in Indonesia are: guava leaves, curry leaves, basil leaves, turmeric leaves, mint leaves, papaya leaves, betel leaves, soursop leaves, aloe vera, and green tea [3]. From the results of a literature study, we found that 80% of people depend on medicinal plants for health, but there are still many people who do not properly utilize the properties of medicinal plants due to incorrect identification of the types of medicinal plants [4]. Identification of medicinal plants can be done by applying classifier technology, i.e., machine learning technology, to make it easier to find out information on types of medicinal plants. Identification of types of medicinal plants can be done by classifying images of medicinal plants using the Convolutional Neural Network (CNN) algorithm.

CNN have demonstrated excellent performance in various image classification tasks. Architectures such as ResNet18 [5], VGG16 [4], and EfficientNet_B0 [6] have been proven to produce high classification accuracy in the image classification domain. However, using a single model is not enough to achieve optimal results because each model has its own strengths and weaknesses [7]. The ensemble

*Corresponding author: nova@utb-univ.ac.id (N. Agustina)

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approach combines several machine learning models to make better image predictions and has been proven in several studies. Ensemble techniques such as average voting are one of the ensemble methods chosen in several case studies because they are proven to produce better performance than other ensemble methods [8].

Research on the identification and classification of medicinal plants using machine learning technology has shown significant progress in recent years. Convolutional Neural Networks (CNN) have become a mainstream method in image classification due to their ability to extract complex image features and provide accurate results [9]. Models such as ResNet18, VGG16, and EfficientNet_B0 are some of the CNN architectures that have been widely used and proven effective in various image classification studies [10]. ResNet18, with its residual learning technique, addresses the vanishing gradient problem, while VGG16 offers a deep yet simple network structure. On the other hand, EfficientNet_B0 combines efficiency and accuracy through a balanced scaling approach between the depth, width, and resolution of the network in the image [11].

In addition to the use of a single CNN model, ensemble techniques have been identified as an effective approach to improve classification performance [12]. Ensemble techniques combine multiple models to make better and more stable predictions. Some popular ensemble methods include voting, averaging, and stacking. The average voting technique, for example, combines predictions from several models by taking the average of the predicted probabilities. This can improve overall accuracy by reducing the variability and errors of individual models [13]. Research has shown that a combination of CNN models can significantly improve classification accuracy compared to a single model [14]. The application of ensemble techniques to CNN models has also shown promising results in various application domains, including medical and object recognition. In the context of medical classification, research has shown that ensemble techniques can improve the accuracy of brain tumor classification by combining ResNet18 and VGG models [15]. The use of ensemble techniques in the classification of medicinal plants, although still relatively rare, has great potential for application considering the morphological diversity of medicinal plants and the need for high accuracy in identifying plant types [16]. This study aims to apply ensemble techniques to the CNN model to classify medicinal plants in Indonesia, which is expected to increase the accuracy and reliability of classification results.

In previous research, CNN models have been applied to various classification tasks with satisfactory results. For example, research has shown that ensemble models can improve medical image classification performance by reducing prediction errors through a combination of powerful models [17]. This approach leverages multiple CNN models to capture different patterns in the data and combine them to produce more accurate and reliable predictions. Ensemble techniques have also been applied in object recognition and image classification, showing superior results compared to the use of a single model [18]. Combining predictions from various models provides advantages in terms of stability and robustness of the classification system [19]. This research uses an ensemble approach combining ResNet18, VGG16, and EfficientNet_B0 with average voting to improve the accuracy of medicinal plant classification in Indonesia. The technique enhances model performance across diverse conditions and data variations, which is crucial due to the high diversity in plant morphology and characteristics.

This research aims to apply ensemble techniques to CNN models like ResNet18, VGG16, and EfficientNet_B0 for improving classification accuracy in identifying medicinal plants in Indonesia. Unlike previous studies, no similar research was found combining ensemble methods with Indonesian medicinal plant datasets. This approach aims to create a more accurate and robust classification system to support plant conservation and usage. The study details the ensemble model development, training, validation, performance evaluation, challenges encountered, and offers recommendations for future research. In summary, the contributions made by this research are:

1. Evaluating the ResNet18, VGG16, and EfficientNet_B0 models, and the ensemble using a representative dataset of Indonesian medicinal plants.
2. Modify output layers of CNN for Implementation of an ensemble method, i.e., average voting, which combines the ResNet18, VGG16, and EfficientNet_B0 models to increase the accuracy of classification of medicinal plants in Indonesia.

II. METHOD

In this section, we describe the methods used to achieve the research objectives, i.e., classifying images of medicinal plants using one of the ensemble methods, the average voting

ensemble. The method is divided into several stages, starting from collecting data on Indonesian medicinal plants, data preprocessing, model design and implementation, model training and validation, to model evaluation. Fig. 1 outlines the steps we carried out in this research to obtain medicinal plant classification results using the ensemble method.

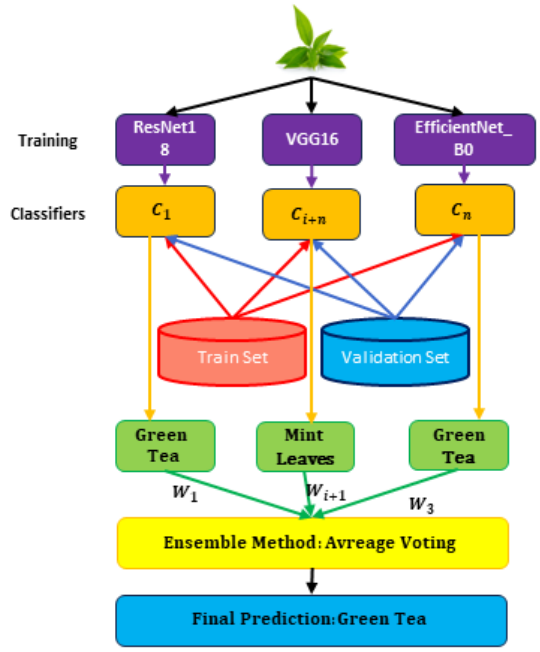


Figure 1. Ensemble method for medicinal plant classification

A. Medicinal Plants Dataset

We determined a dataset of Indonesian medicinal plants called the Herbal Plant Dataset, sourced from Kaggle, which was accessed on May 20, 2024. The dataset consists of 800 training data and 200 testing data. The types of Indonesian medicinal plants available in the dataset are: guava leaves, curry leaves, basil leaves, turmeric leaves, mint leaves, papaya leaves, betel leaves, soursop leaves, aloe vera, and green tea. Table 1 is an example of types of Indonesian medicinal plants sourced from the dataset.

B. Dataset Preprocessing Data

Data preprocessing is carried out by transforming the data to be implemented into a Convolutional Neural Network (CNN) model. The transformations carried out are resizing (changing the size) of the image to a size of 256x256 pixels and converting the image to a tensor. Next, the dataset is divided into training and validation subsets for model training and evaluation purposes. Dataset division is carried out to separate the data that will be used to train the model from the data that will be used to validate the model's performance during training.

Table 1. Medicinal Plants and Their Pharmacological Properties

Image	Name	Medicinal Properties
	Guava Leaves	Antidiabetic, anticancer, anti-inflammatory, antioxidant, antibacterial and antiviral [20]
	Curry Leaves	Analgesic, anticancer, antioxidant, antimicrobial, antihypertensive, antidiabetic, anti-inflammatory [21]
	Basil Leaves	Antioxidant, antihyperglycemic, antidiabetic, anti-inflammatory, antiviral [22]
	Turmeric Leaves	Anti-inflammatory, antioxidant, antibacterial, antitumor, hepatoprotective [23, 24]
	Mint Leaves	Antimicrobial, antioxidant, anti-inflammatory, analgesic, antifungal [25]
	Papaya Leaves	Anticancer, anti-inflammatory, antidiabetic, antiulcer, antioxidant [26]
	Betel Leaves	Antibacterial, antioxidant, anti-inflammatory, anticancer, antiulcer [27]
	Soursop Leaves	Anticancer, antioxidant, anti-inflammatory, antidiabetic [28]
	Alove Vera	Anti-inflammatory, antidiabetic, antibacterial, wound healing, hepatoprotective [29]
	Green Tea	Antioxidant, anticancer, anti-inflammatory, cardioprotective [30]

C. Implementation of Ensemble Models

This study tested the sensitivity of ResNet18, VGG16, and EfficientNet_B0 models on a medicinal plant dataset to evaluate their performance. ResNet18 uses residual blocks to overcome the vanishing gradient problem in deep networks, VGG16 features a deep yet simple architecture known for effective feature extraction, and EfficientNet_B0 scales depth, width, and resolution for improved efficiency and performance. All three models share the same output layer for classification results [31], which is obtained from calculating Equation (1).

$$y = W * x + b \quad (1)$$

ResNet18, VGG16, and EfficientNet_B0 use a fully connected (FC) layer in the output layer to produce the final prediction (y). The final predicti on is obtained from the predetermined weight (W) multiplied by the input value from the previous layer (x) plus the bias value (b). After each model is trained separately, an ensemble method is applied using average voting to combine the predictions from each model. The average voting method is used to combine the probability predictions from each model by taking the average of these probabilities, thus producing a more accurate final prediction. To produce classification results from applying the ensemble, we modify the output layer as shown in Equation (2).

$$y = \frac{1}{N}\sum_{i=1}^{N} y_i \quad (2)$$

In Eq. (2), the voting average (y) is obtained from 1 divided by the number of models in the ensemble (N); in this case study, N = 3. Next, the result is multiplied by the total probability value from the 1st model to the 3rd model in the ensemble for each prediction.

D. Training

At this stage, the medicinal plant dataset is split into training, validation, and testing subsets. Pretrained ResNet18, VGG16, and EfficientNet_B0 models with modified output layers are used. A loss function and optimizer with a 0.001 learning rate and 0.9 momentum are applied for training across 10 epochs. After each epoch, validation accuracy is monitored, and once training is complete, the models are evaluated on the testing dataset for performance metrics, including accuracy, precision, recall, and F1-score.

E. Validation

At the end of each epoch, the model undergoes validation by switching to evaluation mode, ensuring no gradient updates. Predictions are generated from the validation subset and compared to actual labels. Correct predictions are counted to calculate validation accuracy, which is recorded to monitor performance and detect issues like overfitting or underfitting during training.

F. Model Evaluation

The evaluation in this research uses accuracy, precision, recall, and F1-Score to assess model performance. Accuracy measures the proportion of correct predictions, precision focuses on the accuracy of positive predictions, recall indicates the model's ability to detect actual positive samples, and F1-Score balances precision and recall. These metrics are applied to individual models (ResNet18, VGG16, and EfficientNet_B0) and an ensemble model using soft voting, providing a comparison of performance in classifying medicinal plants in Indonesia. The formulas for calculating these metrics are provided in Equation (3) to Equation (6) [32].

$$Accuracy = \frac{TP+TN}{Tp+TN+FP+FN} \quad (3)$$

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

$$Recall = \frac{TP}{TP+FN} \quad (5)$$

$$F1\text{-}Score = \frac{2 \times Precision \times Recall}{Precision+Recall} \quad (6)$$

III. RESULT AND DISCUSSION

A. Result

The average-voting ensemble effectively fuses the class probabilities from ResNet18, VGG16, and EfficientNet_B0 to produce consistent, reliable predictions. EfficientNet_B0 is the strongest individual model, but the ensemble improves precision and stabilizes outcomes by compensating for VGG16's weaker performance while retaining accuracy close to the best single model. This approach delivers a robust classifier for medicinal-plant recognition and is well suited for deployment.

The average voting ensemble combines the outputs of ResNet18, VGG16, and EfficientNet_B0 and yields strong, consistent performance. Based on the evaluation in Fig. 2, EfficientNet_B0 is the best single model with 94.00% accuracy, 94.49% precision, 94.00% recall, and 94.01% F1-Score. ResNet18 attains 87.00% accuracy, 91.16% precision, 87.00% recall, and 87.35% F1-Score. VGG16 is the weakest, with 63.50% accuracy, 63.89% precision, 63.50% recall, and 61.10% F1-Score. The ensemble achieves 93.00% accuracy, 94.99% precision, 93.00% recall, and 93.22% F1-Score, slightly below EfficientNet_B0 in accuracy but the highest precision overall. These results confirm that while EfficientNet_B0 delivers the top standalone accuracy, the ensemble provides a robust alternative with excellent precision and balanced metrics for medicinal-plant classification. These results are visualized in Fig. 2.

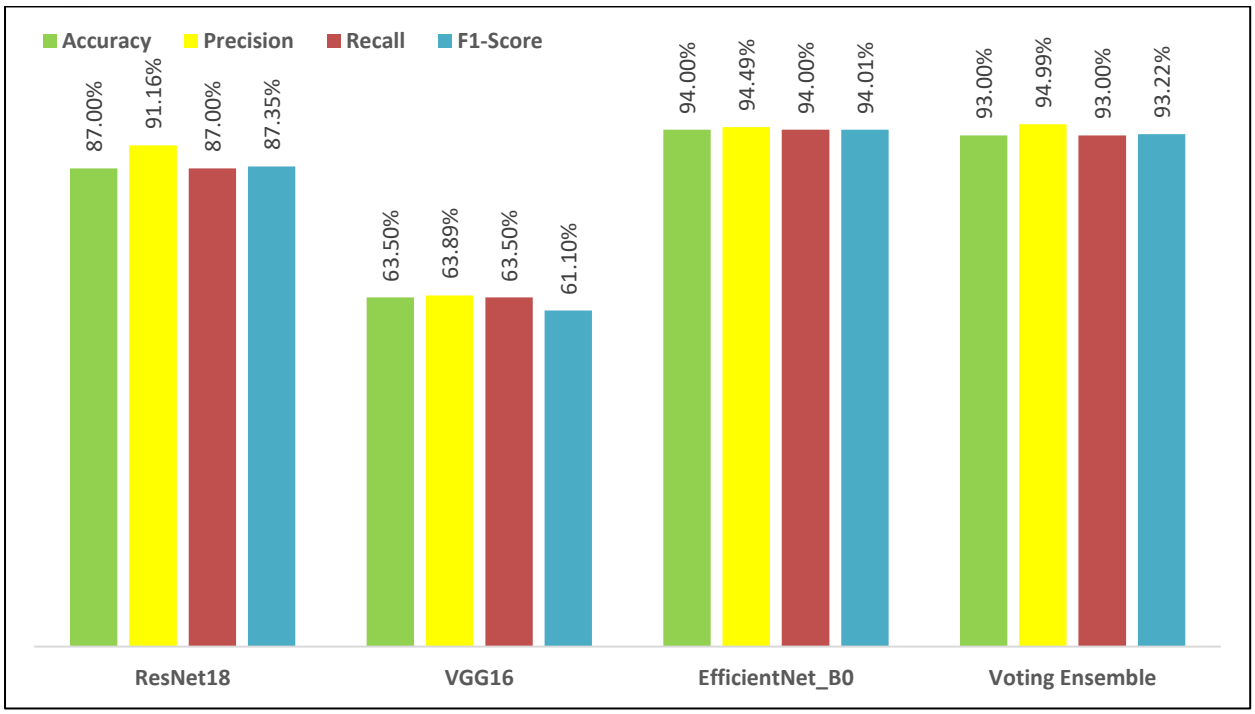


Figure 2. Comparison of Individual Models and Ensemble Method for Medicinal Plant Classification Performance

This study evaluates the model's performance on 200 medicinal-plant images. The ResNet18 model produced 174 correct predictions. Six classes were classified perfectly, i.e., Guava Leaf, Curry Leaf, Basil Leaf, Turmeric Leaf, Mint Leaf, and Soursop Leaf, each correct for 20 images. The main errors occurred in visually similar pairs: Green Tea was often predicted as Basil Leaf for 10 images and as Turmeric Leaf for 1 image. Mint Leaf was frequently predicted as Basil Leaf for 6 images. Aloe Vera was sometimes predicted as Basil Leaf for 5 images and as Green Tea for 2 images. In addition, Basil Leaf was occasionally misclassified as Green Tea for 1 image and as Guava Leaf for 1 image. These test results are summarized in the confusion matrix shown in Fig. 3.

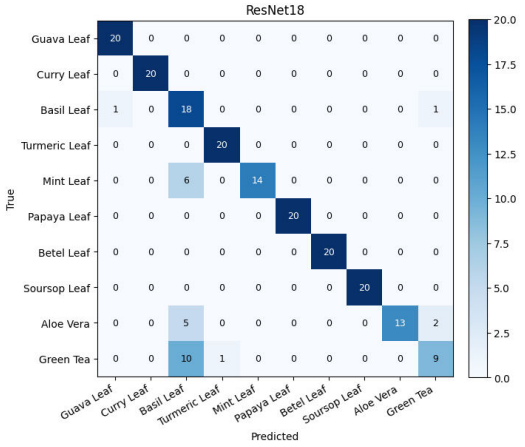


Figure 3. Confusion Matrix Classification ResNet18 Models of Medicinal Plants.

The VGG16 model produced 130 correct predictions (65.0 percent). Two classes were perfectly classified, i.e., Curry Leaf and Soursop Leaf, each correct for 20 images. Errors were broadly distributed across the remaining classes, with the most frequent confusions including Mint Leaf predicted as Papaya Leaf for 14 images, Basil Leaf predicted as Green Tea for 9 images, and Aloe Vera predicted as Basil Leaf for 7 images and as Green Tea for 4 images. These test results are summarized in the confusion matrix shown in Fig. 4.

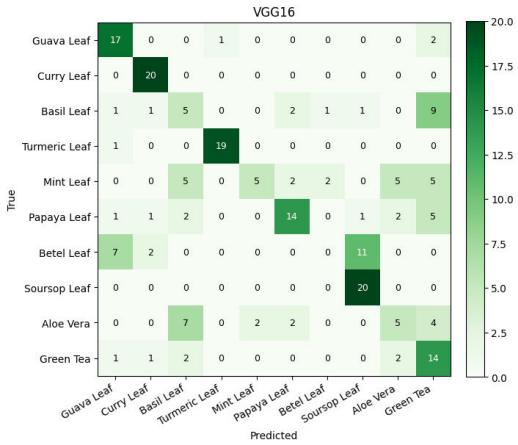


Figure 4. Confusion Matrix Classification VGG16 Models of Medicinal Plants.

The EfficientNet_B0 model produced 188 correct predictions. The EfficientNet_B0 model produced 188 correct predictions, equal to 94.0 percent. Most classes were classified nearly perfectly. Guava Leaf, Curry Leaf, Betel Leaf,

and Soursop Leaf were all correct for 20 of 20 images. Basil Leaf was correct for 19 of 20 images. Turmeric Leaf was correct for 20 of 20 images, with two cases where Basil Leaf was predicted as Turmeric Leaf. Mint Leaf and Papaya Leaf were each correct for 18 of 20 images, with confusions between Mint Leaf and Turmeric Leaf in two images and between Papaya Leaf and Mint Leaf in two images. The main errors occurred in visually similar pairs involving Aloe Vera and Green Tea and the Basil Leaf versus Turmeric Leaf pair. Aloe Vera was predicted as Basil Leaf in four images, as Turmeric Leaf in one image, and as Green Tea in four images. Green Tea was predicted as Basil Leaf in four images and as Turmeric Leaf in one image. These test results are summarized in the confusion matrix shown in Fig. 5.

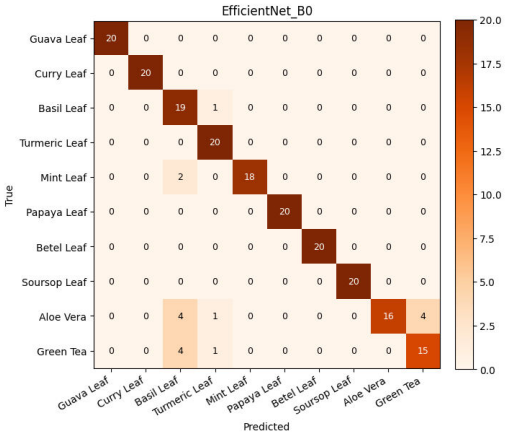


Figure 5. Confusion Matrix Classification EfficienNet_B0 Models of Medicinal Plants.

The ensemble model produced 186 correct predictions. Seven classes were classified perfectly, i.e., Guava Leaf, Curry Leaf, Basil Leaf, Turmeric Leaf, Papaya Leaf, Betel Leaf, and Soursop Leaf, each correct for 20 images. These test results are summarized in the confusion matrix shown in **Fig. 6.**

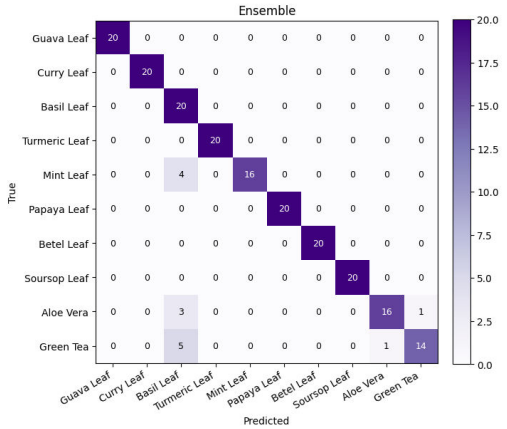


Figure 6. Confusion Matrix Classification Ensemble Models of Medicinal Plants.

Furthermore, the ensemble model remaining errors were concentrated in a few pairs Mint Leaf was predicted as Basil Leaf for 4 images, Aloe Vera was predicted as Curry Leaf for 3 images and as Green Tea for 1 image, and Green Tea was predicted as Guava Leaf for 5 images.

A. Discussion

The results indicate that architecture plays a major role in representation quality. ResNet18 leverages residual blocks that preserve gradient flow through shortcut connections, allowing low-level features to be propagated to higher layers. This mechanism yields stable representations of edges, textures, and leaf venation, which is reflected in consistent predictions across most classes. VGG16 is effective at extracting fine details through repeated small kernels and a well-structured depth, but its errors are dispersed among visually similar classes, leading to uneven robustness across classes. EfficientNet_B0 applies compound scaling to depth, width, and resolution, providing high representational capacity. Under the present training setup, this model requires more systematic augmentation curation and hyperparameter tuning to fully exploit its capacity, particularly for highly similar class pairs such as Aloe Vera versus Green Tea and Basil versus Green Tea.

The ensemble approach with average voting combines the strengths of the three architectures while suppressing their individual weaknesses. ResNet18 contributes stability on texture patterns, VGG16 adds sensitivity to local details, and EfficientNet_B0 offers better generalization to variations in shape and scale. This combination produces more consistent predictions across classes and reduces errors in frequently confused pairs. The impact is direct for field applications, since the system becomes more reliable under variations in illumination, camera viewpoint, and suboptimal leaf conditions.

The main limitations of this study lie in dataset size and diversity, which do not fully capture natural variation, and in a deliberately simple tuning process. Future work will focus on expanding the image corpus across seasons and locations, employing augmentations that emphasize discriminative textures among similar classes, balancing class distributions, and exploring class-weighted or focal loss. Incorporating leaf segmentation before classification, staged fine-tuning at higher resolutions, and probability calibration are also projected to improve decision quality. In practical terms, this ensemble-based design is ready to be directed toward mobile or embedded platforms to

support rapid identification of medicinal plants and biodiversity management in Indonesia. For reproducibility, the full training pipeline, configuration, and plotting scripts are available in our open Kaggle notebook https://www.kaggle.com/code/novaagustina/ensemble-medical-plant/notebook.

IV. CONCLUSION

This study shows that an average-voting ensemble of ResNet18, VGG16, and EfficientNet_B0 delivers robust and consistent performance for medicinal-plant classification in Indonesia. EfficientNet_B0 stands out as the strongest single model, while the ensemble stabilizes predictions by compensating for VGG16's weaker results and leveraging ResNet18's residual learning and VGG16's fine-grained feature extraction. These complementary strengths yield a dependable classifier suitable for real-world use and supportive of biodiversity conservation efforts, where rapid and accurate species recognition is critical.

Further research should include refining the ensemble strategy (e.g., calibrated or weighted voting, stacking), exploring additional and more diverse architectures, and expanding the dataset to cover broader species variation and capture conditions. Further gains are expected from stronger data augmentation, class balancing, loss re-weighting, and deployment-oriented steps such as probability calibration and model compression for mobile or embedded platforms.

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